MATH329 · OPERATIONS RESEARCH

Lecture 4 Simplex Method

Midterm summary · Chapter 3 · the highest-mark question

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What's inside

01 The idea and the two starting requirements

02 How to build the starting tableau

03 The four rules

04 The lecture example in full

05 A min problem (midterm 462)

06 The question's shape in the midterm and its marking

07 Special cases in the simplex method

08 Common mistakes

09 Test yourself

01 The idea and the two starting requirements

Simplex is a tableau on which we repeat the same steps. Each iteration moves us from one corner to a better one, and we stop when nothing can improve. It is the last question in every midterm and carries the most marks: 9 to 15.

Appeared in the midterm (462)

The question: what two requirements must be met to start the simplex algorithm? The answer, word for word:

1. The coefficients of the basic variables (slack variables) form an **Identity matrix**.
2. In the objective function row, the coefficients of the basic variables are **zero**.

Both hold automatically when every constraint is `≤` with a nonnegative right-hand side, because the slack variables give the identity matrix directly.

02 How to build the starting tableau

Maximize $z = 8x_1 + 9x_2$
subject to $2x_1 + 3x_2 \leq 50$
$2x_1 + 6x_2 \leq 80$
$3x_1 + 3x_2 \leq 70$
$x_1, x_2 \geq 0$

1 Standard form

$$\text{Maximize } z = 8x_1 + 9x_2 + 0s_1 + 0s_2 + 0s_3$$

$$\begin{aligned} \text{subject to} \quad & 2x_1 + 3x_2 + s_1 = 50 \\ & 2x_1 + 6x_2 + s_2 = 80 \\ & 3x_1 + 3x_2 + s_3 = 70 \\ & x_1, x_2, s_1, s_2, s_3 \geq 0 \end{aligned}$$

2 Move the objective to one side

$$z - 8x_1 - 9x_2 - 0s_1 - 0s_2 - 0s_3 = 0$$

This is why the objective coefficients enter the Z row with reversed signs: 8 becomes -8.

3 Fill in the tableau

Basic	x_1	x_2	s_1	s_2	s_3	Sol.	Ratio
Z	−8	−9	0	0	0	0	
s_1	2	3	1	0	0	50	$\frac{50}{3}$
s_2	2	**6**	0	1	0	80	$\frac{40}{3}$
s_3	3	3	0	0	1	70	$\frac{70}{3}$

The **Basic** column holds the slack variables and the **Sol.** column holds the right-hand sides. The current solution is $x_1 = x_2 = 0$ with $z = 0$.

03 The four rules

#	Rule	Explanation
1	**Entering variable** Optimality condition	For max: the variable with the most negative coefficient in the `Z` row. For min: the most positive coefficient.
2	**Leaving variable** Feasibility condition	Divide the **Sol.** column by the entering column, only for rows with a positive coefficient. The smallest result leaves. This is the **minimum ratio test**.
3	**Pivot element**	The number where the entering column (**pivot column**) meets the leaving row (**pivot row**).
4	**Row operations**	Make the pivot element 1 and the rest of its column 0, with the two formulas below.

```
New pivot row = Current pivot row ÷ Pivot element
```

```
New row = (Current row) − (Its pivot column coefficient) × (New pivot row)
```

The second formula applies to every other row, including the `Z` row. In the **Basic** column, replace the leaving variable's name with the entering variable's name.

When do we stop?

Type	The solution is optimal when
Max	No negative number is left in the `Z` row.
Min	No positive number is left in the `Z` row.

Summary of the Simplex Method

Step 1. Determine a starting basic feasible solution.

Step 2. Select an entering variable using the optimality condition. Stop if there is no entering variable; the last solution is optimal. Else, go to step 3.

Step 3. Select a leaving variable using the feasibility condition.

Step 4. Determine the new basic solution by using the appropriate Gauss-Jordan computations. Go to step 2.

04 The lecture example in full

We continue with the same problem. The gold cell is the pivot element, and the shaded

column is the entering column.

Basic	x_1	x_2	s_1	s_2	s_3	Sol.	Ratio
Z	−8	−9	0	0	0	0	
s_1	2	3	1	0	0	50	$\frac{50}{3}$
s_2	2	**6**	0	1	0	80	$\frac{40}{3}$
s_3	3	3	0	0	1	70	$\frac{70}{3}$
Z	−5	0	0	$\frac{3}{2}$	0	120	
s_1	**1**	0	1	$-\frac{1}{2}$	0	10	10
x_2	$\frac{1}{3}$	1	0	$\frac{1}{6}$	0	$\frac{40}{3}$	40
s_3	2	0	0	$-\frac{1}{2}$	1	30	15
Z	0	0	5	−1	0	170	
x_1	1	0	1	$-\frac{1}{2}$	0	10	—
x_2	0	1	$-\frac{1}{3}$	$\frac{1}{3}$	0	10	30
s_3	0	0	−2	$\frac{1}{2}$	1	10	20
Z	0	0	1	0	2	190	
x_1	1	0	−1	0	1	20	
x_2	0	1	1	0	$-\frac{2}{3}$	$\frac{10}{3}$	
s_2	0	0	−4	1	2	20	

1 Iteration 1

The most negative entry in the Z row is -9, so x_2 enters. The ratios are $\frac{50}{3} = 16.7$, $\frac{80}{6} = 13.3$ and $\frac{70}{3} = 23.3$. The smallest is 13.3, so s_2 leaves and the pivot is 6.

$$\text{New } x_2 \text{ row} = (2, 6, 0, 1, 0 \mid 80) \div 6 = (\tfrac{1}{3}, 1, 0, \tfrac{1}{6}, 0 \mid \tfrac{40}{3})$$

$$\text{New } Z \text{ row} = Z \text{ row} - (-9) \times \text{new } x_2 \text{ row} = (-5, 0, 0, \tfrac{3}{2}, 0 \mid 120)$$

$$\text{New } s_1 \text{ row} = s_1 \text{ row} - 3 \times \text{new } x_2 \text{ row} = (1, 0, 1, -\tfrac{1}{2}, 0 \mid 10)$$

$$\text{New } s_3 \text{ row} = s_3 \text{ row} - 3 \times \text{new } x_2 \text{ row} = (2, 0, 0, -\tfrac{1}{2}, 1 \mid 30)$$

2 Iteration 2

A -5 remains in the Z row, so x_1 enters. The ratios are $\frac{10}{1} = 10$, $\frac{\frac{40}{3}}{\frac{1}{3}} = 40$ and $\frac{30}{2} = 15$. The smallest is 10, so s_1 leaves and the pivot is 1.

3 Iteration 3

A -1 remains under s_2, so s_2 enters. The first row has a negative coefficient, so it is skipped. The ratios are $\frac{\frac{10}{1}}{\frac{1}{3}} = 30$ and $\frac{\frac{10}{1}}{\frac{1}{2}} = 20$. So s_3 leaves and the pivot is $\frac{1}{2}$.

All entries in the Z row are ≥ 0, so this is the optimal solution.

Optimum $z = 190$ at $x_1 = 20$, $x_2 = \frac{10}{3}$

Check: $z = 8(20) + 9(\frac{10}{3}) = 160 + 30 = 190$

The solution is read from the **Sol.** column: each variable in the **Basic** column has its value beside it, and any variable not in that column is zero.

05 A min problem (midterm 462)

Min Z =	$x_1 - 2x_2$
S.to	$2x_1 + 3x_3 \leq 1$
	$3x_1 + 2x_2 - x_3 \leq 5$
	$x_1, x_2, x_3 \geq 0$

The only difference from max: the entering variable is the most positive entry in the `Z` row, and we stop when no positive entry is left. The `Z` row is built the same way: $z - x_1 + 2x_2 = 0$.

Basic	x_1	x_2	x_3	s_1	s_2	Sol.	Ratio
Z	−1	2	0	0	0	0	
s_1	2	0	3	1	0	1	–
s_2	3	**2**	−1	0	1	5	$\frac{5}{2}$
Z	−4	0	1	0	−1	−5	
s_1	2	0	**3**	1	0	1	$\frac{1}{3}$
x_2	$\frac{3}{2}$	1	$-\frac{1}{2}$	0	$\frac{1}{2}$	$\frac{5}{2}$	–
Z	$-\frac{14}{3}$	0	0	$-\frac{1}{3}$	−1	$-\frac{16}{3}$	
x_3	$\frac{2}{3}$	0	1	$\frac{1}{3}$	0	$\frac{1}{3}$	
x_2	$\frac{11}{6}$	1	0	$\frac{1}{6}$	$\frac{1}{2}$	$\frac{8}{3}$	

Optimum $z = -\frac{16}{3} = -5.33$ at $x_1 = 0$, $x_2 = \frac{8}{3} = 2.67$, $x_3 = \frac{1}{3} = 0.33$

In the first tableau the `s1` row has a zero under `x2`, so it has no ratio. The answer key writes the numbers as decimals: `-4.67`, `1.83`, `0.17`. Both forms are fine.

06 The question's shape in the midterm and its marking

Midterm	What was asked	Marks
472	Standard form, then simplex stopping at the first iteration (two tables only).	5 + 10
471	Standard form, then full simplex for a max problem with two constraints.	–
462	Standard form, then full simplex for a min problem with three variables.	10
461	Standard form, then simplex stopping at the first iteration (two tables only).	9

How it is marked (written in the answer key)

First table 2m (every row 0.5m + 0.5m for pivot element).
Second and third table 3m (1m for every row). Every mistake in calculation −0.5m.

So every row carries a mark and every arithmetic slip costs half a mark. Write every row even if you are unsure of one, and circle the pivot element.

If it says "Only two tables"

Write the starting tableau, do one iteration, write the second tableau and stop. Do not continue to the optimum.

07 Special cases in the simplex method

#	Case	How to spot it in the tableau
1	**Degeneracy**	A tie for the smallest ratio when choosing the leaving variable. The result: a basic variable equal to zero; we may reach a new BFS without improving `z`, and in some cases this leads to **cycling**.
2	**Alternative optima**	In the optimal tableau, a nonbasic variable has a zero coefficient in the `Z` row.
3	**Unbounded solution**	A variable can enter, but its whole column is zero or negative, so no variable can leave.
4	**Infeasible solution**	The constraints are inconsistent. It can never occur when all constraints are `≤` with nonnegative right-hand sides.

1 · Degeneracy (Example 3.5-1)

Maximize $z = 3x_1 + 9x_2$ s.t. $x_1 + 4x_2 \le 8$, $x_1 + 2x_2 \le 4$, $x_1, x_2 \ge 0$

Basic	x_1	x_2	s_1	s_2	Sol.	Ratio
Z	−3	−9	0	0	0	
s_1	1	**4**	1	0	8	2
s_2	1	2	0	1	4	2
Z	$-\frac{3}{4}$	0	$\frac{9}{4}$	0	18	
x_2	$\frac{1}{4}$	1	$\frac{1}{4}$	0	2	8
s_2	**$\frac{1}{2}$**	0	$-\frac{1}{2}$	1	0	0
Z	0	0	$\frac{3}{2}$	$\frac{3}{2}$	18	
x_2	0	1	$\frac{1}{2}$	$-\frac{1}{2}$	2	
x_1	1	0	−1	2	0	

In the first tableau the two ratios are equal: $\frac{8}{4} = 2$ and $\frac{4}{2} = 2$. After the iteration $s_2 = 0$ although it is basic. In the second iteration the solution changes but z stays at 18 with no improvement. The final solution is $z = 18$ at $x_1 = 0$, $x_2 = 2$.

The slide prints $x_2 = 6$ in the last tableau; the correct value is 2. Check: $9 \times 2 = 18$.

2 · Alternative optima (Example 3.5-2)

Maximize $z = 2x_1 + 4x_2$ s.t. $x_1 + 2x_2 \le 5$, $x_1 + x_2 \le 4$, $x_1, x_2 \ge 0$

Basic	x_1	x_2	s_1	s_2	Sol.	Ratio
Z	−2	−4	0	0	0	
s_1	1	2	1	0	5	$\frac{5}{2}$
s_2	1	1	0	1	4	4
Z	0	0	2	0	10	
x_2	$\frac{1}{2}$	1	$\frac{1}{2}$	0	$\frac{5}{2}$	5
s_2	$\frac{1}{2}$	0	$-\frac{1}{2}$	1	$\frac{3}{2}$	3
Z	0	0	2	0	10	
x_2	0	1	1	−1	1	
x_1	1	0	−1	2	3	

The second tableau is optimal: $z = 10$ at $x_2 = \frac{5}{2}$. But x_1 is nonbasic with a zero coefficient in the Z row, so entering it gives another solution with the same value: $z = 10$ at $x_1 = 3$, $x_2 = 1$.

3 · Unbounded solution (Example 3.5-3)

Maximize $z = 2x_1 + x_2$ s.t. $x_1 - x_2 \le 10$, $2x_1 \le 40$, $x_1, x_2 \ge 0$

Basic	x_1	x_2	s_1	s_2	Sol.	Ratio
Z	−2	−1	0	0	0	
s_1	**1**	−1	1	0	10	10
s_2	2	0	0	1	40	20
Z	0	−3	2	0	20	
x_1	1	−1	1	0	10	–
s_2	0	**2**	−2	1	20	10
Z	0	0	−1	$\frac{3}{2}$	50	
x_1	1	0	0	$\frac{1}{2}$	20	–
x_2	0	1	−1	$\frac{1}{2}$	10	–

In the last tableau s_1 has -1 in the Z row, so it should enter. But its column is 0 and -1, with no positive number to divide by. So z can grow without limit.

Appeared in the midterm (471)

If all the ratios in simplex table are negative or infinite then the solution an alternative. (False: the solution is **unbounded**.)

08 Common mistakes

Mistake	Correct
Writing the `Z` row with the objective's own signs	The signs reverse: $8x_1 + 9x_2$ becomes -8 and -9.
Computing a ratio for a row with a negative or zero coefficient	Ratios are only for positive coefficients.
Including the `Z` row in the ratio test	The `Z` row is not part of the test.
Choosing the most negative entry in a min problem	For min the entering variable is the most positive entry.
Forgetting to change the name in the Basic column	The leaving variable is replaced by the entering one.
Continuing after "Only two tables"	Stop after the second tableau.

09 Test yourself

1. What are the two requirements to start simplex?
2. How is the entering variable chosen for max and for min?
3. What does it mean when no variable can leave?
4. Solve by simplex: $\text{Max } z = 3x_1 + 2x_2$ subject to $x_1 + x_2 \leq 4$ and $x_1 \leq 2$.

Answers

#	Answer
1	The coefficients of the basic variables form an identity matrix, and their coefficients in the objective function row are zero.
2	Max: the most negative entry in the `Z` row. Min: the most positive.
3	**Unbounded solution**.
4	`z = 10` at `x₁ = 2`, `x₂ = 2`. The tableaus are below.

Basic	x_1	x_2	s_1	s_2	Sol.	Ratio
Z	−3	−2	0	0	0	
s_1	1	1	1	0	4	4
s_2	**1**	0	0	1	2	2
Z	0	−2	0	3	6	
s_1	0	**1**	1	−1	2	2
x_1	1	0	0	1	2	–
Z	0	0	2	1	10	
x_2	0	1	1	−1	2	
x_1	1	0	0	1	2	